

```
type: SNMP
interval: 5m
```

```
# Trigger expression (95th percentile over
24 h window)
{Template Net Generic SNMP:net.if.in.
last(24h,95p)}>0.7*{$IF_SPEED}
```

Result: a single spike never pages you at midnight again.

Dynamic “Floating” thresholds in prometheus

Prometheus can compare a metric to its rolling mean plus two standard deviations – perfect for catching DNS-query explosions without hard-coding a number.

```
expr: rate(dns_requests_total[5m])
      > (avg_over_time(rate(dns_requests_
total[5m])[1h]) * 2)
for: 2m
labels:
  severity: warning
```

If requests double versus the last hour’s average for two minutes, Alertmanager fires – no manual tuning.



Auto-suppress “Downstream Echo” noise

High-latency WAN often triggers a storm of VoIP, DNS and SaaS alarms. In Zabbix 6.4 create a dependency so Wan-Latency-High suppresses child alerts until cleared. Now you fix the pipe, not 30 symptoms.

Monthly “Noise-audit” ritual

Run `zabbix_sender -r | awk '$2>120 {print}'` to list triggers firing >120 times a month. Either retune thresholds or kill the check – no one reads 300 Slack pings.

Troubleshooting: Slow Wi-Fi, DNS lag & cabling faults

According to Jones IT, as much as 80% of “network is slow” tickets in SMBs map back to only three root causes: RF congestion, DNS mis-hits, or layer-1 cable errors. Here’s a 15-minute triage drill that nails each.

Jones IT found mirrored glass partitions and cordless phones top India’s interference list – switch to 6 GHz Wi-Fi 6E or 5 GHz DFS to escape.

DNS Latency: Two-minute diagnosis

1. `dig +trace google.com` – if root to .in is fast (<30 ms) but final answer is 400 ms, your resolver is slow.
2. Point `/etc/resolv.conf` to 1.1.1.1; retest. Fixed? Change router DNS.
3. Still slow? Check `systemd-resolved --statistics` for cache misses >60 %. Possibly forwarder loop.



Large cache or unpatched forwarder bugs often cause 300 ms lookup stalls. Clear cache nightly: `rndc flush`.

Cabling faults: No more guesswork

A simple TDR-enabled tester like the TREND Networks LanTEK

WI-FI WOES IN FIVE STEPS

Step	Command / Tool	Pass/Fail Hint
1. Check client RSSI	<code>iw dev wlan0 link</code>	<-70 dBm? relocate or add AP
2. Scan for co-channel APs	<code>sudo wavemon</code>	>5 APs on same 6 GHz channel – change channel
3. Look for retries	Zabbix item wlan.tx.retry	>15 %? interference or low power
4. Test throughput	<code>iperf3 -c server -P4</code>	<40 % of PHY rate → airtime jam
5. Validate roam	<code>sudo nmcli dev wifi list</code>	Sticky client? Lower AP min-RSSI